

中国石油大学（华东）第三十一届高等数学竞赛答案

一、（本题满分 20 分）

1. (10 分) 设函数 $f(x)$ 连续, 且 $f(0) \neq 0$, 求极限 $\lim_{x \rightarrow 0} \frac{\int_0^x (x-t)f(t)dt}{x \int_0^x f(x-t)dt}$.

解 由于 $\int_0^x f(x-t)dt \stackrel{x-t=u}{=} \int_x^0 f(u)(-du) = \int_0^x f(u)du$, 于是

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^x (x-t)f(t)dt}{x \int_0^x f(x-t)dt} &= \lim_{x \rightarrow 0} \frac{x \int_0^x f(t)dt - \int_0^x tf(t)dt}{x \int_0^x f(u)du} \\ &= \lim_{x \rightarrow 0} \frac{\int_0^x f(t)dt + xf(x) - xf(x)}{\int_0^x f(u)du + xf(x)} = \lim_{x \rightarrow 0} \frac{\int_0^x f(t)dt}{\int_0^x f(u)du + xf(x)} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\int_0^x f(t)dt}{x}}{\frac{\int_0^x f(u)du}{x} + f(x)} = \frac{f(0)}{f(0) + f(0)} = \frac{1}{2}. \end{aligned}$$

2. (10 分) 设 $x \in (0,1)$, 试证明: $(1+x)\ln^2(1+x) < x^2$.

证明: 令 $f(x) = (1+x)\ln^2(1+x) - x^2$,

因为 $f'(x) = \ln^2(1+x) + 2\ln(1+x) - 2x$,

$$f''(x) = 2 \frac{\ln(1+x)+1}{x+1} - 2, \quad f'''(x) = -2 \frac{\ln(1+x)}{(1+x)^2},$$

又由 Taylor 公式可知, $\exists \xi \in (0, x)$,

$$\text{使 } f(x) = f(0) + f'(0)x + \frac{1}{2!} f''(0)x^2 + \frac{1}{3!} f'''(\xi)x^3 = -\frac{\ln(1+\xi)}{3(1+\xi)^2} x^3.$$

$$\text{因为 } \xi \in (0, x) \in (0, 1) \Rightarrow -\frac{\ln(1+\xi)}{3(1+\xi)^2} < 0,$$

$$\therefore f(x) < 0, \text{ 即 } (1+x)\ln^2(1+x) < x^2.$$

二、（本题满分 20 分）

1. (10 分) 证明: $\frac{1}{\frac{1}{1 \times 2018} + \frac{1}{2018 \times 4035} + \frac{1}{4035 \times 6052} + \frac{1}{6052 \times 8069} + \dots} = 2017$.

证明: 设 $x_1 = 2018$

$$x_2 = x_1(2x_1 - 1) = 2018 \times 4035$$

$$x_3 = (2x_1 - 1)(3x_1 - 2) = 4035 \times 6052 \dots$$

$$x_n = [(n-1)x_1 - (n-2)][nx_1 - (n-1)], \dots$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{x_1} + \left(\frac{2}{2x_1 - 1} - \frac{1}{x_1} \right) + \left(\frac{3}{3x_1 - 2} - \frac{2}{2x_1 - 1} \right) + \dots + \left(\frac{n}{nx_1 - (n-1)} - \frac{n-1}{(n-1)x_1 - (n-2)} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{n}{nx_1 - (n-1)} = \lim_{n \rightarrow \infty} \frac{1}{x_1 - \frac{n-1}{n}} = \frac{1}{x_1 - 1} = \frac{1}{2017}$$

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = 2017$$

2. (10 分) 设 $f(x)$ 是定义在 $(0, +\infty)$ 上的单调增加连续函数, 证明: 对任何 $b > a > 0$ 成立不等式

$$\int_a^b xf(x)dx \geq \frac{1}{2} [b \int_0^b f(x)dx - a \int_0^a f(x)dx].$$

证明: 设 $F(x) = x \int_0^x f(t)dt$, $x \geq 0$, 则

$$b \int_0^b f(x)dx - a \int_0^a f(x)dx = F(b) - F(a) = \int_a^b F'(x)dx$$

$$= \int_a^b \left[\int_0^x f(t)dt + xf(x) \right] dx \leq \int_a^b \left[\int_0^x f(x)dt + xf(x) \right] dx$$

$$= \int_a^b [xf(x) + xf(x)]dx = 2 \int_a^b xf(x)dx$$

$$\text{所以 } \int_a^b xf(x)dx \geq \frac{1}{2} \left[b \int_0^b f(x)dx - a \int_0^a f(x)dx \right].$$

三、(本题满分 20 分)

$$1. (10 \text{ 分}) \text{ 计算 } I = \int_1^2 dx \int_{\sqrt{x-1}}^1 \frac{\sin y}{y} dy$$

$$\text{解 令 } F(x) = \int_{\sqrt{x-1}}^1 \frac{\sin y}{y} dy, \text{ 则 } I = \int_1^2 F(x)dx = xF(x) \Big|_1^2 - \int_1^2 xF'(x)dx$$

$$= - \int_0^1 \frac{\sin y}{y} dy + \int_1^2 x \frac{\sin \sqrt{x-1}}{\sqrt{x-1}} \cdot \frac{dx}{2\sqrt{x-1}}$$

$$= - \int_0^1 \frac{\sin x}{x} dx + \frac{1}{2} \int_1^2 \frac{x}{x-1} \cdot \sin \sqrt{x-1} dx$$

$$\text{而 } \int_1^2 \frac{x}{x-1} \cdot \sin \sqrt{x-1} dx \stackrel{\sqrt{x-1}=y}{=} \int_0^1 \frac{y^2+1}{y^2} \sin y \cdot 2y dy$$

$$= 2 \int_0^1 y \sin y dy + 2 \int_0^1 \frac{\sin y}{y} dy$$

$$\text{故 } I = \int_0^1 y \sin y dy = -y \cos y \Big|_0^1 + \int_0^1 \cos y dy = \sin 1 - \cos 1$$

$$2. (10 \text{ 分}) \text{ 求积分 } I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin x + x^2}{(x \cos x)^2} dx.$$

$$\text{解 } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin x + x^2}{(x \cos x)^2} dx \text{ 为广义积分, 瑕点为 } x=0$$

$$\text{对于 } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x^2}{(x \cos x)^2} dx = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x dx = 2$$

$$\text{对于 } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin x}{(x \cos x)^2} dx, \text{ 由于 } \lim_{x \rightarrow 0} x f(x) = \lim_{x \rightarrow 0} x \frac{\sin x}{(x \cos x)^2} = 1, \text{ 根据广义积分判别法则知}$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin x}{(x \cos x)^2} dx \text{ 发散, } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin x + x^2}{(x \cos x)^2} dx \text{ 发散.}$$

四、(本题满分 20 分)

$$1. (10 \text{ 分}) \text{ 设函数 } y = y(x) \text{ 由参数方程 } \begin{cases} x = 1 + 2t^2, \\ y = \int_1^{1+2\ln t} \frac{e^u}{u} du \end{cases} (t > 1) \text{ 所确定, 求 } \frac{d^2 y}{dx^2} \Big|_{x=9}$$

$$\text{解 由 } \frac{dy}{dt} = \frac{e^{1+2\ln t}}{1+2\ln t} \cdot \frac{2}{t} = \frac{2et}{1+2\ln t}, \quad \frac{dx}{dt} = 4t, \text{ 得 } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2et}{4t(1+2\ln t)} = \frac{e}{2(1+2\ln t)},$$

$$\text{所以 } \frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{1}{\frac{dx}{dt}} = \frac{e}{2} \cdot \frac{-1}{(1+2\ln t)^2} \cdot \frac{2}{t} \cdot \frac{1}{4t} = -\frac{e}{4t^2(1+2\ln t)^2}.$$

当 $x=9$ 时, 由 $x=1+2t^2$ 及 $t>1$ 得 $t=2$, 故

$$\frac{d^2 y}{dx^2} \Big|_{x=9} = -\frac{e}{4t^2(1+2\ln t)^2} \Big|_{t=2} = -\frac{e}{16(1+2\ln 2)^2}.$$

$$2. (10 \text{ 分}) \text{ 计算 } \int_0^1 dx \int_0^x dy \int_0^y \frac{\sin z}{1-z} dz$$

$$\text{解 令原式} = \int_0^1 f(x) dx, \text{ 其中 } f(x) = \int_0^x dy \int_0^y \frac{\sin z}{1-z} dz = \int_0^x g(y) dy, \quad g(y) = \int_0^y \frac{\sin z}{1-z} dz$$

$$\begin{aligned}
 f(1) &= \int_0^1 g(y) dy = y g(y) \Big|_0^1 - \int_0^1 \frac{y \sin y}{1-y} dy = g(1) - \int_0^1 \frac{y \sin y}{1-y} dy \\
 &= \int_0^1 \frac{\sin y}{1-y} dy - \int_0^1 \frac{y \sin y}{1-y} dy = \int_0^1 \sin y dy = -\cos y \Big|_0^1 = 1 - \cos 1
 \end{aligned}$$

又因为 $f'(x) = g(x)$, 所以

$$\begin{aligned}
 \int_0^1 f(x) dx &= x f(x) \Big|_0^1 - \int_0^1 x g(x) dx = f(1) - \int_0^1 x g(x) dx = f(1) - \frac{1}{2} \int_0^1 g(x) dx^2 \\
 &= f(1) - \frac{1}{2} x^2 g(x) \Big|_0^1 + \frac{1}{2} \int_0^1 x^2 g'(x) dx = f(1) - \frac{1}{2} \int_0^1 \frac{\sin x}{1-x} dx + \frac{1}{2} \int_0^1 \frac{x^2 \sin x}{1-x} dx \\
 &= f(1) - \frac{1}{2} \int_0^1 \frac{(1-x^2) \sin x}{1-x} dx = f(1) - \frac{1}{2} \int_0^1 (1+x) \sin x dx = \frac{1}{2} (1 - \sin 1)
 \end{aligned}$$

五、(本题满分 20 分)

1. (10 分) 计算 $\iint_{\Sigma} |xyz| dS$, Σ 为锥面 $x^2 + y^2 = z^2$ 位于 xOy 上方, 平面 $z=1$ 下方的部分。

解 设第一象限内的部分为 Σ_1 : $x \geq 0, y \geq 0, \sqrt{x^2 + y^2} = z \leq 1$

$$\begin{aligned}
 \iint_{\Sigma} |xyz| dS &= \iint_{D_{xy}} |xyz| \sqrt{1 + \frac{\partial z^2}{\partial x} + \frac{\partial z^2}{\partial y}} dx dy dz = 4 \iint_{D_{xy}} |xyz| \sqrt{1 + 4x^2 + 4y^2} dx dy \\
 &= 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \sin \theta \cos \theta \cdot r^2 \cdot \sqrt{1 + 4r^2} r dr = 4 \left(\frac{1}{2} \sin^2 \theta \Big|_0^{\frac{\pi}{2}} \right) \cdot \left(\frac{1}{2} \int_0^1 r^4 \sqrt{1 + 4r^2} dr^2 \right) \\
 \sqrt{1 + 4r^2} &= u \int_1^{\sqrt{5}} u \cdot \left(\frac{u^2 - 1}{4} \right)^2 = \int \frac{2u^2}{4^3} (u^2 - 1)^2 du = \frac{125\sqrt{5} - 1}{420}
 \end{aligned}$$

2. (10 分) 求 $I = \int_L xy ds$, L : 椭圆 $\begin{cases} x = a \cos t, \\ y = b \sin t, \end{cases}$ (第 I 象限)。

$$\begin{aligned}
 \text{解: } I &= \int_0^{\frac{\pi}{2}} a \cos t \cdot b \sin t \sqrt{(-a \sin t)^2 + (b \cos t)^2} dt \\
 &= ab \int_0^{\frac{\pi}{2}} \sin t \cos t \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} dt, \quad \text{令 } u = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} \\
 du &= \frac{2a^2 \sin t \cos t - 2b^2 \cos t \sin t}{2\sqrt{a^2 \sin^2 t + b^2 \cos^2 t}} dt = \frac{\sin t \cos t (a^2 - b^2)}{\sqrt{a^2 \sin^2 t + b^2 \cos^2 t}} dt. \\
 \text{则 } I &= \frac{ab}{a^2 - b^2} \int_b^a u^2 du = \frac{ab(a^2 + ab + b^2)}{3(a + b)}
 \end{aligned}$$